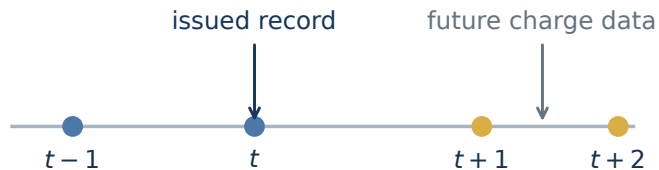
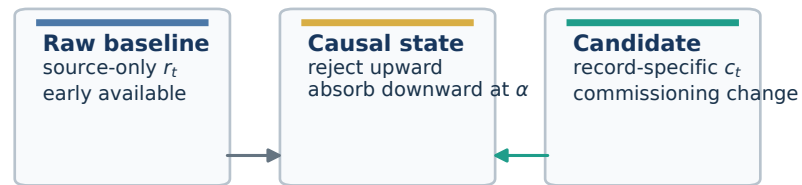


A Deployment failure: future data revise the past



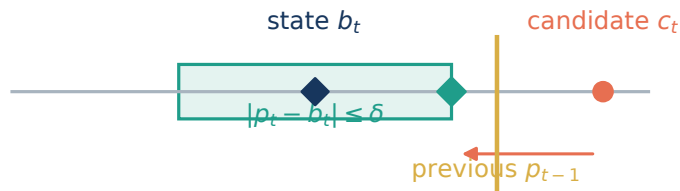
Online requirement: the prefix already emitted at t must remain invariant

B Prefix-causal state and learned proposal



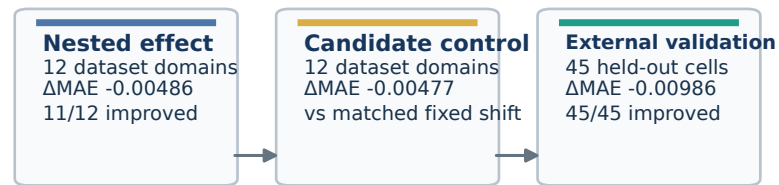
$b_t = b_{t-1} + \alpha \min(r_t - b_{t-1}, 0)$ uses only the observed prefix

C Recursive feasible interval and exact harm tube



$$p_t = \Pi_{[\max(0, b_t - \delta), \min(1.3, b_t + \delta, p_{t-1})]}(c_t)$$

D Evidence chain: distinct questions, distinct units



All three 95% intervals excluded zero; all deterministic certificates held